

Module 4: Intermediate Statistics and Inference — Answer Key

Prequiz

1. The interval 165 to 195 is $\mu \pm 1\sigma$, so about 68%. The value $210 = 180 + 2(15)$, so about 2.5% lie above it: 2.5%.
The middle 95% is 180 ± 30 , or 150 g to 210 g.

2.

$$z_A = \frac{520 - 500}{100} = \text{0.20}, \quad z_B = \frac{28 - 21}{5} = \text{1.40}.$$

Student B has the stronger relative performance.

3.

$$243 \pm 8 = [\text{235}, \text{251}].$$

The claim 230 is not inside the interval. The value 243 is a **sample statistic**.

4. To halve M , multiply n by 4:

$$4(200) = \text{800}.$$

5.

$$z = \frac{78 - 60}{12} = \text{1.5}.$$

Yes, it is within two standard deviations. The z-score describes an **individual observation** relative to the distribution.

Matching quiz

1. 19 to 51 is $35 \pm 16 = \mu \pm 2\sigma$, so about 95%.

2.

$$z = \frac{84 - 72}{6} = \text{2}.$$

The score is above the mean.

3.

$$47\% \pm 6\% = [\text{41\%}, \text{53\%}].$$

The value 55% is not inside the interval.

4. 31.2 cm is a **sample statistic**; it estimates the population mean fish length.

5. Multiplying n by 9 multiplies $\sqrt{n}$ by 3, so the margin of error is multiplied by 1/3.